

Follow and create algorithms involving a sequence of steps and decisions to investigate numbers; describe any emerging patterns

Learning intention

- follow and create algorithms involving a sequence of steps and decisions to investigate numbers; describe any emerging patterns.

Teach and model

- following or creating an algorithm that determines whether a given number is a multiple of 2, 5 or 10, identifying and discussing emerging patterns.
- creating a sorting algorithm that will sort a collection of 5 cent and 10 cent coins and providing the total value of the collection by applying knowledge of multiples of 5 and 10.
- following or creating an algorithm to generate number patterns formed by doubling and halving using technology to assist where appropriate; identifying and describing emerging patterns.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.